Test Case Document (updated after Sprint 1 and Sprint 2)

Project	Online Exam Portal (Software Engineering mini project)
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This document expands the representative test case IDs in the Software Test Plan (§13 RTM) into full test cases, and updates them for what changed during the sprints. Where a sprint changed the behaviour, the **Sprint changes** row says what changed and why; the reasons are also in `deviations.md`. **Actual result** and **Status** come from the latest test run.

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Test case	Requirement	Title	Type	Priority	Status
TC-Auth-01	OEP-F-001	Student registers with a valid email or roll number	Functional	High	Passed
TC-Auth-02	OEP-F-002	User logs in with username, email or roll number	Functional	High	Passed
TC-Auth-03	OEP-F-003	Each role reaches only its own screens and APIs	Functional	High	Passed
TC-Auth-04	OEP-F-004	Five consecutive failed logins lock the account	Functional	Medium	Passed
TC-QB-01	OEP-F-010	Faculty creates questions of each type	Functional	High	Passed
TC-QB-02	OEP-F-011	Questions are tagged and can be filtered	Functional	Medium	Passed
TC-QB-03	OEP-F-012	Faculty imports questions from the template	Functional	Medium	Passed
TC-QB-04	OEP-F-013	Edit lock on questions in use	Functional	Medium	Passed
TC-Exam-01	OEP-F-020	Faculty assembles an exam manually and by rules	Functional	High	Passed
TC-Exam-02	OEP-F-021	Exam scoring settings are saved and validated	Functional	High	Passed
TC-Exam-03	OEP-F-022	Exam opens and closes on its window	Functional	High	Passed
TC-Exam-04	OEP-F-023	Each student gets an independently shuffled paper	Functional	Medium	Passed

Test case	Requirement	Title	Type	Priority	Status
TC-Exam-05	OEP-F-024	Faculty assigns an exam to a course and publishes it	Functional	High	Not executed
TC-Deliver-01	OEP-F-030	Exam screen with server-synced timer	Functional	High	Passed
TC-Deliver-02	OEP-F-031	Navigator and flag for review	Functional	Medium	Passed
TC-Deliver-03	OEP-F-032	Attempt is submitted automatically on timeout	Functional	High	Passed
TC-Deliver-04	OEP-F-033	Student submits early	Functional	High	Passed
TC-Deliver-05	OEP-F-034	Single-attempt rule and re-attempt allowance	Functional	High	Passed
TC-Deliver-06	OEP-F-035	Answers are auto-saved and survive a disconnect	Functional	Medium	Passed
TC-Eval-01	OEP-F-040	Objective answers are graded on submission	Functional	High	Passed
TC-Eval-02	OEP-F-041	Pending written answers appear in the grading queue	Functional	High	Passed
TC-Eval-03	OEP-F-042	Faculty grades a written answer	Functional	Medium	Passed
TC-Eval-04	OEP-F-043	Final attempt score calculation	Functional	High	Passed
TC-Result-01	OEP-F-050	Student sees their scorecard	Functional	High	Passed
TC-Result-02	OEP-F-051	Faculty/Admin view the class report	Functional	Medium	Passed
TC-Result-03	OEP-F-052	Faculty export the report	Functional	Medium	Passed
TC-Result-04	OEP-F-053	Faculty view percentage correct per question	Functional	Low	Passed
TC-Admin-01	OEP-F-060	Admin manages Faculty and Student accounts	Functional	High	Passed
TC-Admin-02	OEP-F-061	Admin creates courses and maps Faculty/Students	Functional	Medium	Passed
TC-Admin-03	OEP-F-062	Admin retrieves the audit log	Functional	Medium	Passed
TC-Admin-04	OEP-F-063	Admin checks system health	Functional	Low	Passed
TC-Perf-01	OEP-NF-001	200 concurrent exam starts, P90 ≤ 2 s	Non-functional	High	Not executed
TC-Rel-01	OEP-NF-002	Saved responses survive client loss	Non-functional	High	Passed
TC-Ops-01	OEP-NF-003	Uptime during a simulated exam window	Non-functional	High	Passed
TC-Audit-01	OEP-NF-004	Audit logging and retention	Non-functional	Medium	Passed
TC-UX-01	OEP-NF-005	Usability and accessibility of the exam website	Non-functional	Medium	Partly passed
TC-Arch-01	OEP-NF-006	Architecture review against the SAD	Non-functional	Medium	Partly passed
TC-Sec-01	OEP-SR-001	TLS enforcement	Security	High	Passed

Test case	Requirement	Title	Type	Priority	Status
TC-Sec-02	OEP-SR-002	Password hashing	Security	High	Passed
TC-Sec-03	OEP-SR-003	Session expiry and invalidation	Security	High	Passed
TC-Sec-04	OEP-SR-004	Answer-key confidentiality	Security	High	Passed
TC-Sec-05	OEP-SR-005	Lockout audit trail	Security	Medium	Passed
TC-Sec-06	OEP-SR-006	Server-side role check on every endpoint	Security	High	Passed

TC-Auth-01

Field	Value
Requirement	OEP-F-001: Student self-registration with institutional email or roll number
Title	Student registers with a valid email or roll number
Type / level	Functional / Integration (API)
Priority	High
Preconditions	Portal running; no account exists for the email or roll number.
Steps	1. POST /api/auth/register/ with first name, email and/or roll number and a password meeting the policy. 2. Repeat with an email that already exists. 3. Repeat with a weak password. 4. Repeat with role=ADMIN in the body.
Test data	student@pes.edu / PES2UG25CS999, password Str0ng-Passphrase!
Expected result	201 and a Student account is created; duplicate email/roll number gives 409; weak password gives 422; the role field is ignored and the account is always a Student.
Sprint changes	SAD's single `name` field split into first and last name; `role` in the request is ignored so nobody can self-register as Faculty/Admin; errors keep DRF's field-error format instead of the SAD envelope.
Automated tests	`accounts/tests.py::StudentRegistrationTests` `accounts/tests.py::UserModelTests`
Other evidence	—
Actual result	17 of 17 automated tests passed.
Status	**Passed**

TC-Auth-02

Field	Value
Requirement	OEP-F-002: Login checks credentials against stored hashes
Title	User logs in with username, email or roll number
Type / level	Functional / Integration (API)
Priority	High
Preconditions	Active Student, Faculty and Admin accounts exist.
Steps	1. POST /api/auth/login/ with email + correct password. 2. Repeat with roll number, then username. 3. Repeat with a wrong password. 4. Call GET /api/auth/me/ with the returned token, then POST /api/auth/logout/.
Test data	Seeded by the test (see automated tests)
Expected result	Valid credentials return 200 with {token, role, user, expires_in}; wrong password returns 400 with a generic message; /me returns the user; after logout the token no longer works.
Sprint changes	Website (web app) logs in with a Django session through the same backend instead of an API token.

Automated tests	`accounts/test_login.py::LoginTests`
	`accounts/test_login.py::TokenSessionTests`
	`web/tests.py::LoginPageTests`

Other evidence	—
Actual result	24 of 24 automated tests passed.
Status	**Passed**

TC-Auth-03

Field	Value
Requirement	OEP-F-003: Role-based access for Admin, Faculty and Student
Title	Each role reaches only its own screens and APIs
Type / level	Functional / Integration (API)
Priority	High

Field	Value
Preconditions	One account per role.
Steps	1. For every /api/ route, call it anonymously and as each role. 2. Student opens a Faculty-only endpoint (e.g. question bank). 3. Faculty opens an Admin-only endpoint.
Test data	Seeded by the test (see automated tests)
Expected result	Anonymous gets 401, wrong role gets 403, allowed role succeeds; no route is reachable without an explicit rule.
Sprint changes	Temporary is_staff checks from Sprint 1 replaced by role permissions (accounts.permissions).
Automated tests	`accounts/test_rbac.py::EndpointRegistryTests` `accounts/test_rbac.py::RoleMatrixTests`
Other evidence	—
Actual result	9 of 9 automated tests passed.
Status	**Passed**

TC-Auth-04

Field	Value
Requirement	OEP-F-004: Lock account after five failed logins and log the event
Title	Five consecutive failed logins lock the account
Type / level	Functional / Integration (API)
Priority	Medium
Preconditions	Active Student account.
Steps	1. Send 5 logins with a wrong password. 2. Send a 6th login with the correct password. 3. Wait past the lock duration (time moved forward in the test) and log in.
Test data	Seeded by the test (see automated tests)
Expected result	After the 5th failure the account is locked; correct password returns 423 while locked; login works after 15 minutes and the failure count resets.
Sprint changes	SRS says 'temporarily'; lock duration fixed at 15 minutes (ACCOUNT_LOCKOUT_DURATION).
Automated tests	`accounts/test_lockout.py::AccountLockoutTests`

Field	Value
Other evidence	—
Actual result	12 of 12 automated tests passed.
Status	**Passed**

TC-QB-01

Field	Value
Requirement	OEP-F-010: Faculty add MCQ, True/False and Short-Answer questions
Title	Faculty creates questions of each type
Type / level	Functional / Integration (API)
Priority	High
Preconditions	Logged in as Faculty.
Steps	1. POST /api/questions/ with an MCQ (4 options, answer among them). 2. Create a True/False and a Short-Answer question. 3. Try an MCQ with 1 option, duplicate options, or an answer not in the options.
Test data	Seeded by the test (see automated tests)
Expected result	Valid questions return 201 and are listed; invalid ones return 400 with the field error.
Sprint changes	Fill-in-the-blank added as a fourth question type so it can be auto-graded (see TC-Eval-01).
Automated tests	`question_bank/tests.py::QuestionAPITests` `question_bank/tests.py::QuestionModelTests`
Other evidence	—
Actual result	18 of 18 automated tests passed.
Status	**Passed**

TC-QB-02

Field	Value
Requirement	OEP-F-011: Tag questions by subject, topic and difficulty

Field	Value
Title	Questions are tagged and can be filtered
Type / level	Functional / Integration (API)
Priority	Medium
Preconditions	Faculty; questions with different tags exist.
Steps	1. Create a question without subject/topic/difficulty. 2. GET /api/questions/?subject=...&topic=...&difficulty=...
Test data	Seeded by the test (see automated tests)
Expected result	Missing or invalid tags return 400; filters return only matching questions, and filters combine.
Sprint changes	None. Implemented as specified.
Automated tests	`question_bank/tests.py::QuestionAPITests`
Other evidence	—
Actual result	8 of 8 automated tests passed.
Status	**Passed**

TC-QB-03

Field	Value
Requirement	OEP-F-012: Bulk import from CSV/Excel template
Title	Faculty imports questions from the template
Type / level	Functional / Integration (API)
Priority	Medium
Preconditions	Faculty; template downloaded from /api/questions/template/.
Steps	1. Upload a valid CSV, then a valid .xlsx. 2. Upload a file with one bad row. 3. Upload a file with a missing header, an unsupported type, or over the size limit.
Test data	Seeded by the test (see automated tests)
Expected result	Valid files create all rows; a bad row rejects the whole file with the row number and saves nothing; bad files are rejected with a clear message.
Sprint changes	None. Implemented as specified.

Field	Value
Automated tests	`question_bank/tests.py::QuestionBulkImportTests` `question_bank/tests.py::QuestionTemplateDownloadTests`
Other evidence	—
Actual result	12 of 12 automated tests passed.
Status	**Passed**

TC-QB-04

Field	Value
Requirement	OEP-F-013: Edit/delete questions unless attached to a live or published exam
Title	Edit lock on questions in use
Type / level	Functional / Integration (API)
Priority	Medium
Preconditions	Faculty; one question unused, one in a draft exam, one in a published/live exam, one with a student response.
Steps	1. Edit and delete the unused question. 2. Edit the draft-exam question. 3. Edit/delete the published-exam question. 4. Delete the question with a student response.
Test data	Seeded by the test (see automated tests)
Expected result	Unused and draft-exam questions can be changed; published/live ones and ones with student responses are blocked.
Sprint changes	Also blocks questions that already have student responses (fix in OEP-F-013 PR #82) so past results can't change.
Automated tests	`question_bank/tests.py::QuestionDetailAPITests`
Other evidence	—
Actual result	10 of 10 automated tests passed.
Status	**Passed**

TC-Exam-01

Field	Value
Requirement	OEP-F-020: Assemble an exam by hand-picking or by rules
Title	Faculty assembles an exam manually and by rules
Type / level	Functional / Integration (API)
Priority	High
Preconditions	Faculty; a question bank with Easy/Medium/Hard questions.
Steps	1. POST /api/exams/ to create a draft. 2. Assemble it with a list of question IDs. 3. Generate it by rule (e.g. 2 Easy, 1 Hard). 4. Request more questions than exist.
Test data	Seeded by the test (see automated tests)
Expected result	Draft created; manual order kept; rule picks the requested counts; impossible rule returns 400 and keeps the old questions.
Sprint changes	None. Implemented as specified.
Automated tests	`exam_management/tests.py::ExamAPITests` `exam_management/tests.py::ExamAssemblyTests`
Other evidence	—
Actual result	19 of 19 automated tests passed.
Status	**Passed**

TC-Exam-02

Field	Value
Requirement	OEP-F-021: Set duration, marks, per-question marks and negative marking
Title	Exam scoring settings are saved and validated
Type / level	Functional / Integration (API)
Priority	High
Preconditions	Faculty; draft exam with questions.

Field	Value
Steps	1. Set duration, negative marking and penalty. 2. Set per-question marks and read the total. 3. Try zero duration or zero penalty with negative marking on. 4. Try to change settings after publishing.
Test data	Seeded by the test (see automated tests)
Expected result	Valid settings are saved and total marks are the sum of question marks; invalid values return 400; published exams can't be changed.
Sprint changes	None. Implemented as specified.
Automated tests	`exam_management/tests.py::ExamAPITests`
Other evidence	—
Actual result	8 of 8 automated tests passed.
Status	**Passed**

TC-Exam-03

Field	Value
Requirement	OEP-F-022: Availability window with start and end date/time
Title	Exam opens and closes on its window
Type / level	Functional / Integration (API)
Priority	High
Preconditions	Faculty; draft exam.
Steps	1. Set availability_start and availability_end. 2. Set an end earlier than the start, or only one of the two. 3. As a Student, open the exam before, during and after the window.
Test data	Seeded by the test (see automated tests)
Expected result	Valid window saved; invalid windows return 400; the paper is served only inside the window.
Sprint changes	None. Implemented as specified.
Automated tests	`exam_management/tests.py::ExamAPITests`
Other evidence	—

Field	Value
Actual result	4 of 4 automated tests passed.
Status	**Passed**

TC-Exam-04

Field	Value
Requirement	OEP-F-023: Shuffle question and option order per student
Title	Each student gets an independently shuffled paper
Type / level	Functional / Unit
Priority	Medium
Preconditions	Exam with MCQ and True/False questions.
Steps	1. Build the paper for two different students (seeds). 2. Build it twice for the same student. 3. Grade answers given on a shuffled paper.
Test data	Seeded by the test (see automated tests)
Expected result	Different students can get different question and option orders; the same student always gets the same order; True/False stays True, False; no answer fields; grading is unaffected.
Sprint changes	None. Implemented as specified.
Automated tests	`exam_management/test_shuffle_paper.py::ShufflePaperTests`
Other evidence	—
Actual result	11 of 11 automated tests passed.
Status	**Passed**

TC-Exam-05

Field	Value
Requirement	OEP-F-024: Assign exams to groups/courses and publish/unpublish
Title	Faculty assigns an exam to a course and publishes it

Field	Value
Type / level	Functional / Integration (API)
Priority	High
Preconditions	Faculty; course with mapped students (TC-Admin-02).
Steps	1. Assign the exam to a course. 2. Publish it; a student in the course opens it. 3. A student outside the course tries to open it. 4. Unpublish it.
Test data	Seeded by the test (see automated tests)
Expected result	Only students in the assigned course see and open a published exam; unpublishing hides it.
Sprint changes	Not implemented in Sprints 1–2 (moved to backlog). Until then every open exam with questions is listed for every student, the grading queue is not filtered by faculty, and reports are per exam not per course. Recorded in deviations.md.
Automated tests	—
Other evidence	—
Actual result	Feature not built in Sprints 1–2.
Status	**Not executed**

TC-Deliver-01

Field	Value
Requirement	OEP-F-030: Questions shown with a server-synced countdown timer
Title	Exam screen with server-synced timer
Type / level	Functional / Integration (API)
Priority	High
Preconditions	Student; exam open with a 30-minute duration.
Steps	1. POST /api/exams//start/. 2. Read the timer; change the client clock / reload. 3. Call start again (resume).

Field	Value
Test data	Seeded by the test (see automated tests)
Expected result	Start returns the shuffled questions and remaining_seconds from the server deadline (ends_at); client clock or reload does not change it; resume returns the same attempt with the correct remaining time.
Sprint changes	Website polls the timer every 30 s (the SAD's fallback) instead of a WebSocket. True/False always sent with options True and False.

Automated tests	`exam_delivery/test_delivery.py::ExamDeliveryTimerTests`
	`exam_delivery/test_delivery.py::TrueFalseOptionsTests`
	`web/tests.py::TakeExamPageTests`

Other evidence	—
Actual result	22 of 22 automated tests passed.
Status	**Passed**

TC-Deliver-02

Field	Value
Requirement	OEP-F-031: Flag questions and move freely between them
Title	Navigator and flag for review
Type / level	Functional / Integration (API)
Priority	Medium
Preconditions	Student in an active attempt.
Steps	1. Flag question 2, unflag it, flag question 3. 2. GET the navigator. 3. On the website, use 'Go to a flagged question' and submit.
Test data	Seeded by the test (see automated tests)
Expected result	Flags toggle; navigator shows total, answered and flagged per question; website shows states without relying on colour and warns how many are still flagged on submit.
Sprint changes	Navigator key and jump-to-flagged added in Sprint 2 (OEP-F-031 PR #88).
Automated tests	`exam_delivery/test_delivery.py::QuestionNavigationAndFlagTests` `web/tests.py::TakeExamPageTests`
Other evidence	browser_checks/results.md

Field	Value
Actual result	5 of 5 automated tests passed. Browser check: jump-to-flagged worked and submit warned '1 still flagged'.
Status	**Passed**

TC-Deliver-03

Field	Value
Requirement	OEP-F-032: Auto-submit when time runs out
Title	Attempt is submitted automatically on timeout
Type / level	Functional / Integration (API)
Priority	High
Preconditions	Student in an attempt; time moved past ends_at.
Steps	1. After the deadline, call the timer/autosave/attempt endpoint. 2. Try to save an answer.
Test data	Seeded by the test (see automated tests)
Expected result	Attempt becomes SUBMITTED with reason TIMEOUT and is graded; late answers are ignored / return 410.
Sprint changes	Auto-submit is lazy: it happens when the attempt is next touched, not by a background job. The deadline is still enforced on the server.
Automated tests	`exam_delivery/test_delivery.py::AutoSubmitTimeoutTests`
Other evidence	—
Actual result	2 of 2 automated tests passed.
Status	**Passed**

TC-Deliver-04

Field	Value
Requirement	OEP-F-033: Manual submit before the time limit
Title	Student submits early
Type / level	Functional / Integration (API)
Priority	High

Field	Value
Preconditions	Student in an attempt with answers saved.
Steps	1. POST submit. 2. Try to change an answer afterwards.
Test data	Seeded by the test (see automated tests)
Expected result	Attempt becomes SUBMITTED with reason MANUAL, objective answers are graded immediately, further changes are refused.
Sprint changes	None. Implemented as specified.
Automated tests	`exam_delivery/test_delivery.py::ManualSubmitTests`
Other evidence	—
Actual result	2 of 2 automated tests passed.
Status	**Passed**

TC-Deliver-05

Field	Value
Requirement	OEP-F-034: No second attempt unless allowed by Faculty/Admin
Title	Single-attempt rule and re-attempt allowance
Type / level	Functional / Integration (API)
Priority	High
Preconditions	Student with a submitted attempt.
Steps	1. Start the same exam again. 2. Send two Start requests at the same time on a fresh exam. 3. Faculty allows a re-attempt; student starts again.
Test data	Seeded by the test (see automated tests)
Expected result	Second start returns 403 ExamAlreadyAttempted; simultaneous starts create one attempt; after allowance a new attempt starts.
Sprint changes	Race between simultaneous starts fixed with a DB constraint (one IN_PROGRESS attempt per student and exam).
Automated tests	`exam_delivery/test_delivery.py::ConcurrentStartTests` `exam_delivery/test_delivery.py::SingleAttemptRuleTests`
Other evidence	—

Field	Value
Actual result	7 of 7 automated tests passed.
Status	**Passed**

TC-Deliver-06

Field	Value
Requirement	OEP-F-035: Auto-save at regular intervals
Title	Answers are auto-saved and survive a disconnect
Type / level	Functional / Integration (API)
Priority	Medium
Preconditions	Student in an attempt.
Steps	1. PUT autosave with a batch of answers. 2. Simulate a 7-minute disconnect and resume. 3. On the website: change an answer, go offline, reconnect, reload.
Test data	Seeded by the test (see automated tests)
Expected result	Answers are saved atomically; after reconnect the answers are kept and only the elapsed time is lost; the website saves again on reconnect and shows the last save time.
Sprint changes	SRS says 'regular intervals'; the page saves 2 s after the last change and every 30 s if anything is unsaved, plus on reconnect and when the tab is hidden (Sprint 2, PR #87).
Automated tests	`exam_delivery/test_delivery.py::PeriodicAutosaveTests`
Other evidence	browser_checks/results.md
Actual result	3 of 3 automated tests passed. Browser checks: failed autosave retried and saved; offline then reconnect re-saved and re-synced the timer.
Status	**Passed**

TC-Eval-01

Field	Value
Requirement	OEP-F-040: Auto-grade MCQ, True/False and fill-in-the-blank on submit

Field	Value
Title	Objective answers are graded on submission
Type / level	Functional / Unit
Priority	High
Preconditions	Submitted attempt with objective questions.
Steps	1. Submit correct, wrong, blank and differently-cased answers. 2. Enable negative marking and submit a wrong answer.
Test data	Seeded by the test (see automated tests)
Expected result	Correct = full marks; blank = 0; wrong = −penalty when negative marking is on; comparison ignores case and extra spaces; short-answer is not auto-graded.
Sprint changes	Fill-in-the-blank added; blank answers always score 0 (no penalty).
Automated tests	`evaluation/tests.py::ObjectiveGradingCloseOutTests` `question_bank/test_grading.py`
Other evidence	—
Actual result	16 of 16 automated tests passed.
Status	**Passed**

TC-Eval-02

Field	Value
Requirement	OEP-F-041: Route subjective answers to Faculty
Title	Pending written answers appear in the grading queue
Type / level	Functional / Integration (API)
Priority	High
Preconditions	Submitted attempts with short-answer questions.
Steps	1. Faculty GET /api/evaluation/pending/. 2. Filter by exam_id. 3. Grade one response and reload the queue. 4. Student calls the endpoint.

Field	Value
Test data	Seeded by the test (see automated tests)
Expected result	Queue lists pending responses grouped per exam; graded ones leave the queue; Students get 403.
Sprint changes	No faculty-to-exam link yet (depends on OEP-F-024), so every Faculty/Admin sees all pending responses.
Automated tests	`evaluation/test_grading_queue.py::GradingQueueTests`
Other evidence	—
Actual result	13 of 13 automated tests passed.
Status	**Passed**

TC-Eval-03

Field	Value
Requirement	OEP-F-042: Faculty award marks and feedback on subjective answers
Title	Faculty grades a written answer
Type / level	Functional / Integration (API)
Priority	Medium
Preconditions	Faculty; pending subjective response.
Steps	1. PATCH /api/evaluation/responses// with marks and feedback. 2. Send marks above the maximum or negative. 3. Grade an objective response or one in an unsubmitted attempt. 4. Regrade it.
Test data	Seeded by the test (see automated tests)
Expected result	Marks and feedback are saved and the attempt score is finalised in the same request; invalid marks 400; objective 400; unsubmitted 409; regrade reported as regraded: true.
Sprint changes	None. Implemented as specified.
Automated tests	`evaluation/test_subjective_evaluation.py::EvaluateResponseTests` `evaluation/test_subjective_evaluation.py::FinalScoreIntegrationTests`
Other evidence	—
Actual result	21 of 21 automated tests passed.

Field	Value
Status	**Passed**

TC-Eval-04

Field	Value
Requirement	OEP-F-043: Final score once objective and subjective parts are graded
Title	Final attempt score calculation
Type / level	Functional / Unit
Priority	High
Preconditions	Submitted attempt with objective and subjective questions.
Steps	1. Finalise while a subjective answer is pending. 2. Grade all subjective answers and finalise. 3. Regrade one answer and finalise again. 4. Finalise an objective-only exam and an expired attempt.
Test data	Seeded by the test (see automated tests)
Expected result	Pending answers block finalisation; total = objective + subjective, rounded to 2 dp; regrading recalculates; objective-only exams finalise; expired/in-progress attempts are rejected.
Sprint changes	None. Implemented as specified.
Automated tests	`evaluation/tests.py::FinalizeAttemptScoreTests`
Other evidence	—
Actual result	12 of 12 automated tests passed.
Status	**Passed**

TC-Result-01

Field	Value
Requirement	OEP-F-050: Individual scorecard when evaluation is finished
Title	Student sees their scorecard

Field	Value
Type / level	Functional / Integration (API)
Priority	High
Preconditions	Student with a graded attempt and one still being graded.
Steps	1. GET /api/results// for the graded attempt. 2. Same for the ungraded attempt. 3. Another student opens it.
Test data	Seeded by the test (see automated tests)
Expected result	Graded: total, objective, subjective, max and per-question marks with feedback, no correct answers; ungraded 409; other student 403.
Sprint changes	Scorecard is API-only; there is no website page for it yet.
Automated tests	`evaluation/test_scorecard.py::EndToEndTests` `evaluation/test_scorecard.py::ScorecardTests`
Other evidence	—
Actual result	14 of 14 automated tests passed.
Status	**Passed**

TC-Result-02

Field	Value
Requirement	OEP-F-051: Class/course performance report
Title	Faculty/Admin view the class report
Type / level	Functional / Integration (API)
Priority	Medium
Preconditions	Exam with several graded and one pending attempt.
Steps	1. GET /api/reports/exams//. 2. Student calls the same endpoint.
Test data	Seeded by the test (see automated tests)
Expected result	Average, highest, lowest, 10-bucket distribution and per-student results; pending attempts not counted; Student 403.
Sprint changes	Report is per exam, not per course, because exams aren't linked to courses yet (OEP-F-024).

Field	Value
Automated tests	`reports/test_exam_report.py::ExamReportTests`
Other evidence	—
Actual result	13 of 13 automated tests passed.
Status	**Passed**

TC-Result-03

Field	Value
Requirement	OEP-F-052: Export results as PDF or Excel
Title	Faculty export the report
Type / level	Functional / Integration (API)
Priority	Medium
Preconditions	Exam with graded results.
Steps	1. GET .../export/?export_format=pdf. 2. GET .../export/?export_format=xlsx. 3. Use a student name starting with '='.
Test data	Seeded by the test (see automated tests)
Expected result	Valid PDF and .xlsx files with the same figures as the on-screen report; formula-like text stays text.
Sprint changes	Parameter is export_format because DRF reserves format.
Automated tests	`reports/test_export.py::ExportTests`
Other evidence	—
Actual result	18 of 18 automated tests passed.
Status	**Passed**

TC-Result-04

Field	Value
Requirement	OEP-F-053: Question-wise breakdown

Field	Value
Title	Faculty view percentage correct per question
Type / level	Functional / Integration (API)
Priority	Low
Preconditions	Exam with graded results.
Steps	1. Faculty GET /api/reports/exams//questions/. 2. Admin and Student call it.
Test data	Seeded by the test (see automated tests)
Expected result	Per question: attempts, percentage with full marks, average marks; no answer keys; Admin and Student get 403.
Sprint changes	None. Implemented as specified.
Automated tests	`reports/test_breakdown.py::QuestionBreakdownTests`
Other evidence	—
Actual result	10 of 10 automated tests passed.
Status	**Passed**

TC-Admin-01

Field	Value
Requirement	OEP-F-060: Admin creates, edits and deactivates accounts
Title	Admin manages Faculty and Student accounts
Type / level	Functional / Integration (API)
Priority	High
Preconditions	Logged in as Admin.
Steps	1. Create a Faculty and a Student. 2. Edit a user and reset their password. 3. Deactivate the Student and try to log in as them. 4. Reactivate.
Test data	Seeded by the test (see automated tests)
Expected result	Accounts created/edited; deactivated user can't log in and their token is revoked; records kept; reactivation restores access.

Field	Value
Sprint changes	Admin accounts are managed only via createsuperuser / Django admin so one admin can't lock out another.
Automated tests	`accounts/test_account_management.py::AccountManagementTests`
Other evidence	—
Actual result	26 of 26 automated tests passed.
Status	**Passed**

TC-Admin-02

Field	Value
Requirement	OEP-F-061: Admin manages courses and maps users
Title	Admin creates courses and maps Faculty/Students
Type / level	Functional / Integration (API)
Priority	Medium
Preconditions	Admin; active Faculty and Student accounts.
Steps	1. Create, edit and delete a course. 2. Map a Faculty and Students, then unmap one. 3. Map a user with the wrong role or an inactive user.
Test data	Seeded by the test (see automated tests)
Expected result	CRUD works with unique course codes; mapping/unmapping works; wrong role or inactive users are rejected.
Sprint changes	None. Implemented as specified.

Automated tests	`courses/tests.py::CourseAccessTests`
	`courses/tests.py::CourseCrudTests`
	`courses/tests.py::CourseMappingTests`

Other evidence	—
Actual result	20 of 20 automated tests passed.
Status	**Passed**

TC-Admin-03

Field	Value
Requirement	OEP-F-062: Admin reviews system-wide logs and integrity flags
Title	Admin retrieves the audit log
Type / level	Functional / Integration (API)
Priority	Medium
Preconditions	Audit entries exist (logins, submissions, integrity flags).
Steps	1. Admin GET the audit log, filtered by actor and date range, with limit/offset. 2. Send bad filters. 3. Faculty/Student call it.
Test data	Seeded by the test (see automated tests)
Expected result	Entries returned with filters and paging; bad filters 400; non-admins 403.
Sprint changes	None. Implemented as specified.
Automated tests	`accounts/test_audit_log.py::RetentionTests`
Other evidence	—
Actual result	6 of 6 automated tests passed.
Status	**Passed**

TC-Admin-04

Field	Value
Requirement	OEP-F-063: Diagnostic panel for server and database health
Title	Admin checks system health
Type / level	Functional / Integration (API)
Priority	Low
Preconditions	Admin account.

Field	Value
Steps	1. GET /api/admin/diagnostics/. 2. Open /system-health/ on the website. 3. Simulate a database failure. 4. Student opens both.
Test data	Seeded by the test (see automated tests)
Expected result	Shows server status, DB connectivity and latency, and live counts; a DB failure shows a plain message; Students get 403.
Sprint changes	Website panel added in Sprint 2 (PR #86); refreshes on demand instead of polling.
Automated tests	`exam_delivery/test_diagnostics.py::AdminDiagnosticsTests` `web/tests.py::SystemHealthPanelTests`
Other evidence	—
Actual result	14 of 14 automated tests passed.
Status	**Passed**

TC-Perf-01

Field	Value
Requirement	OEP-NF-001: Everyday actions within 2 s under load (200 concurrent exam-takers)
Title	200 concurrent exam starts, P90 ≤ 2 s
Type / level	Non-functional / System
Priority	High
Preconditions	PostgreSQL database; 200 student accounts; one open exam.
Steps	1. Release 200 concurrent POST /start/ requests together. 2. Record P50/P90/P99 and error rate.
Test data	Seeded by the test (see automated tests)
Expected result	P90 ≤ 2 s and every request returns 201.
Sprint changes	Test written but only runs on PostgreSQL; the project and CI use SQLite, so no measured run yet.
Automated tests	`exam_delivery/test_nfr.py::NonFunctionalRequirementsTests`
Other evidence	performance_report.md

Field	Value
Actual result	0 of 1 automated tests passed, 1 skipped. Load test exists but is skipped on SQLite; no measured run recorded.
Status	**Not executed**

TC-Rel-01

Field	Value
Requirement	OEP-NF-002: No lost responses on dropped connections
Title	Saved responses survive client loss
Type / level	Non-functional / Integration (API)
Priority	High
Preconditions	Student in an attempt with autosaved answers.
Steps	1. Autosave answers, discard the client session, log in again and resume. 2. Submit twice.
Test data	Seeded by the test (see automated tests)
Expected result	All autosaved answers are present after resuming; repeated submit is idempotent.
Sprint changes	Verified by client-session replacement and browser offline/reconnect checks, not a network-layer fault injection.
Automated tests	`exam_delivery/test_nfr.py::NonFunctionalRequirementsTests`
Other evidence	browser_checks/results.md (offline/reconnect, reload/resume)
Actual result	1 of 1 automated tests passed. Browser checks: answers kept after reload and after offline/reconnect.
Status	**Passed**

TC-Ops-01

Field	Value
Requirement	OEP-NF-003: ≥ 99% uptime during scheduled exam windows
Title	Uptime during a simulated exam window
Type / level	Non-functional / System
Priority	High

Field	Value
Preconditions	Server running; uptime probe pointed at /health/.
Steps	1. Run deploy/uptime_probe.py for a 25-minute window while 10 students sit an exam. 2. Stop the database and probe again.
Test data	Seeded by the test (see automated tests)
Expected result	≥ 99% of probes up; /health/ returns 503 when the DB is unreachable.
Sprint changes	SRS measures monthly on a real server; no hosting yet, so a simulated window was used (docs/availability_report.md).

Automated tests	`config/test_health.py::HealthEndpointTests`
	`config/test_health.py::ProbeOnceTests`
	`config/test_health.py::SqliteLockingTests`
	`config/test_health.py::UptimeSummaryTests`
	`exam_delivery/test_nfr.py::NonFunctionalRequirementsTests`

Other evidence	availability_report.md
Actual result	15 of 15 automated tests passed. Simulated 25-minute window with 10 students: 750 of 750 probes up (100%).
Status	**Passed**

TC-Audit-01

Field	Value
Requirement	OEP-NF-004: Key actions logged and kept ≥ 1 academic year
Title	Audit logging and retention
Type / level	Non-functional / Integration (API)
Priority	Medium
Preconditions	Portal with activity.
Steps	1. Log in, submit, evaluate, trigger an integrity flag. 2. Try to edit or delete an audit entry. 3. Run archive_audit_logs (incl. --dry-run) with entries older and newer than the retention period.
Test data	Seeded by the test (see automated tests)
Expected result	Each action creates an entry with actor and time; entries are append-only; only entries past one year are archived and archiving is itself logged; answer content is never logged.

Field	Value
Sprint changes	None. Implemented as specified.

Automated tests	`accounts/test_audit_log.py::AppendOnlyTests`
	`accounts/test_audit_log.py::KeyActionLoggingTests`
	`accounts/test_audit_log.py::RetentionTests`

Other evidence	—
Actual result	23 of 23 automated tests passed.
Status	**Passed**

TC-UX-01

Field	Value
Requirement	OEP-NF-005: First-time student completes an exam unaided; WCAG 2.1 AA
Title	Usability and accessibility of the exam website
Type / level	Non-functional / System
Priority	Medium
Preconditions	Seeded sample exam (docs/browser_checks/seed.py).
Steps	1. Run the automated browser checks (axe-core, keyboard-only, 375 px, offline). 2. Run moderated sessions with 3 first-time students (docs/usability_test.md).
Test data	Seeded by the test (see automated tests)
Expected result	0 axe violations on every main screen; keyboard-only sitting works; ≥ 3 first-time students finish unaided.
Sprint changes	None. Implemented as specified.
Automated tests	`web/tests.py::DashboardTests` `web/tests.py::StaticFilesTests`
Other evidence	browser_checks/results.md, usability_test.md
Actual result	14 of 14 automated tests passed. Automated browser checks: 14 of 14 passed, 0 axe violations on 10 screens. Moderated sessions with first-time students not run yet.
Status	**Partly passed**

TC-Arch-01

Field	Value
Requirement	OEP-NF-006: Horizontal scaling; UI, logic and data layers separated
Title	Architecture review against the SAD
Type / level	Non-functional / Review
Priority	Medium
Preconditions	Code on main.
Steps	1. Review module structure, layering and in-memory state against SAD §3.4–3.7.
Test data	Seeded by the test (see automated tests)
Expected result	Separate UI/service/data modules; no per-user state in application memory; review signed off.
Sprint changes	None. Implemented as specified.
Automated tests	—
Other evidence	architecture_review.md
Actual result	Review: separate modules mostly met, stateless application layer met; awaiting team sign-off.
Status	**Partly passed**

TC-Sec-01

Field	Value
Requirement	OEP-SR-001: TLS 1.2+ for every client-server connection
Title	TLS enforcement
Type / level	Security / System
Priority	High
Preconditions	nginx with deploy/nginx.conf; DJANGO_TLS_ENABLED=1.
Steps	1. Check Django HTTPS settings. 2. Run deploy/tls_check/check_tls.sh: HTTP redirect, HSTS, TLS 1.2/1.3 accepted, TLS 1.0/1.1 refused.
Test data	Seeded by the test (see automated tests)

Field	Value
Expected result	HTTP redirects to HTTPS; HSTS sent; secure cookies; only TLS 1.2 and 1.3 accepted.
Sprint changes	TLS ends at nginx; HTTPS settings are opt-in with DJANGO_TLS_ENABLED so local development stays on HTTP.

Automated tests	`config/test_tls.py::ClientIpTests`
	`config/test_tls.py::ProxiedLoginAuditTests`
	`config/test_tls.py::TlsRequestTests`
	`config/test_tls.py::TlsSettingsTests`

Other evidence	tls_deployment.md (check_tls.sh run on 11-11-2026: 11/11 checks passed)
Actual result	17 of 17 automated tests passed. check_tls.sh against the real nginx.conf: TLS 1.2/1.3 accepted, TLS 1.0/1.1 refused, HTTP→HTTPS redirect, HSTS.
Status	**Passed**

TC-Sec-02

Field	Value
Requirement	OEP-SR-002: Passwords stored only as bcrypt/Argon2 hashes
Title	Password hashing
Type / level	Security / Unit
Priority	High
Preconditions	Any account.
Steps	1. Register a user and read the stored password. 2. Log in with an old PBKDF2 hash. 3. Trigger an error report during login.
Test data	Seeded by the test (see automated tests)
Expected result	Stored value is an Argon2 hash; old hashes upgrade on login; passwords masked in error reports.
Sprint changes	None. Implemented as specified.
Automated tests	`accounts/test_password_hashing.py::PasswordHashingTests` `accounts/test_password_hashing.py::SensitiveParameterTests`
Other evidence	—
Actual result	6 of 6 automated tests passed.

Field	Value
Status	**Passed**

TC-Sec-03

Field	Value
Requirement	OEP-SR-003: Session tokens expire when idle and on logout
Title	Session expiry and invalidation
Type / level	Security / Integration (API)
Priority	High
Preconditions	Logged-in user.
Steps	1. Stay idle past 30 minutes, then call an API. 2. Make requests inside the timeout. 3. Log out and reuse the token.
Test data	Seeded by the test (see automated tests)
Expected result	Idle token returns 401 and is deleted; activity extends the session; logged-out token is rejected.
Sprint changes	Idle timeout fixed at 30 min (TOKEN_IDLE_TIMEOUT_MINUTES); login returns expires_in (snake_case).
Automated tests	`accounts/test_session_expiry.py::IdleTimeoutSettingTests` `accounts/test_session_expiry.py::SessionExpiryTests`
Other evidence	—
Actual result	14 of 14 automated tests passed.
Status	**Passed**

TC-Sec-04

Field	Value
Requirement	OEP-SR-004: Questions and answers never exposed before or during the exam
Title	Answer-key confidentiality
Type / level	Security / Integration (API)

Field	Value
Priority	High
Preconditions	Student; exam before, during and after its window.
Steps	1. Request the paper outside the window. 2. Inspect every response during the exam for correct_answer. 3. Call every endpoint that carries answers as a Student.
Test data	Seeded by the test (see automated tests)
Expected result	Outside the window 403 ExamNotOpen; no response contains correct answers; answer-carrying endpoints return 403.
Sprint changes	None. Implemented as specified.

Automated tests	`exam_management/test_answer_key_protection.py::ExamWindowTests`
	`exam_management/test_answer_key_protection.py::StudentApiExposureTests`
	`exam_management/test_answer_key_protection.py::StudentPaperContentTests`

Other evidence	—
Actual result	15 of 15 automated tests passed.
Status	**Passed**

TC-Sec-05

Field	Value
Requirement	OEP-SR-005: Lockout after five failed logins, logged for audit
Title	Lockout audit trail
Type / level	Security / Integration (API)
Priority	Medium
Preconditions	Active account.
Steps	1. Fail login 5 times. 2. Read the audit log.
Test data	Seeded by the test (see automated tests)
Expected result	One account.locked entry with the account, client IP and lock expiry.
Sprint changes	None. Implemented as specified.

Field	Value
Automated tests	`accounts/test_lockout.py::LockoutAuditTests`
Other evidence	—
Actual result	4 of 4 automated tests passed.
Status	**Passed**

TC-Sec-06

Field	Value
Requirement	OEP-SR-006: RBAC enforced at the server for every endpoint
Title	Server-side role check on every endpoint
Type / level	Security / Integration (API)
Priority	High
Preconditions	One account per role.
Steps	1. Walk every registered /api/ route anonymously and as each role (accounts/test_rbac.py). 2. Compare with docs/endpoint_roles.md.
Test data	Seeded by the test (see automated tests)
Expected result	Every route denies by default and matches the endpoint-to-role list; a new unprotected route fails the test.
Sprint changes	None. Implemented as specified.

Automated tests	`accounts/test_rbac.py::EveryEndpointRoleCheckTests`
	`exam_management/tests.py::ExamAPITests`
	`question_bank/tests.py::QuestionAPITests`
	`question_bank/tests.py::QuestionBulkImportTests`
	`question_bank/tests.py::QuestionDetailAPITests`
	`question_bank/tests.py::QuestionTemplateDownloadTests`

Other evidence	endpoint_roles.md
Actual result	17 of 17 automated tests passed. Every route matches the endpoint-to-role list.
Status	**Passed**